

An in-silico senolytic-enabled regimen for complete periodontal regeneration

The binding axis is lost-bone/cementum neogenesis: a periodontal instance of the universal neogenesis-efficiency bottleneck

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Abstract

We report an in-silico structural result for complete periodontal regeneration — not symptom control but restoration of a chronically periodontitic field to a never-diseased state. Decomposing the diseased periodontium into three cell-state classes by their distance from recoverability — reversible gingiva (mass 0.40, achievable efficiency $\eta = 0.95$), dormant periodontal ligament (PDL, 0.30, $\eta = 0.85$), and lost alveolar bone + cementum (0.30, $\eta = 0.55$) — and driving the two recoverable classes to their achievable optima, the complete-restoration ceiling saturates at $\sum_c m_c \eta_c = 0.80$, *below* a pre-registered ≥ 0.90 -of-normal cure gate (**BLOCK**). The single binding axis is lost-bone/cementum neogenesis: the gate requires $\eta_{\text{bone}} \geq 0.88$, which current guided-tissue-regeneration / enamel-matrix / BMP therapy (≈ 0.80 ceiling) does not reach. Senescent gingival and PDL fibroblasts accumulate in chronic periodontitis and sustain a destructive SASP that suppresses resident osteo-/cemento-progenitor proliferation; modelling a senolytic niche-clearance lever $\eta_{\text{bone}}(\phi) = 0.55 + \phi \cdot 0.45$, the cure ceiling crosses the gate at clearance $\phi \approx 73\text{--}80\%$ ($\eta_{\text{bone}} = 0.88\text{--}0.91$, ceiling $0.90\text{--}0.91$). The regeneration arm is BMP/Wnt/RUNX2 osteo-/cemento-neogenesis run inside the cleared window. This is a concrete instance of the universal neogenesis-bottleneck framework. We claim no efficacy: the η values are literature-order (**ORANGE**); the *structural* result — which axis binds, and that senolytic niche-clearance addresses it — is **GREEN** and robust to the exact inputs. Reproduction: `exports/PERIO-CURE/`.

1 Introduction

Periodontitis is the leading cause of adult tooth loss: chronic bacterial inflammation destroys the tooth-supporting apparatus — gingiva, periodontal ligament (PDL), cementum, and alveolar bone. The clinical goal of *regenerative* periodontal therapy is not to arrest disease but to rebuild the lost attachment apparatus to a healthy state. Guided tissue regeneration (GTR) [9], enamel-matrix derivative (Emdogain) [4], and bone-morphogenetic-protein grafts [12] each recover *part* of the apparatus, but consensus reviews place predictable complete regeneration — new bone, new cementum, and a functionally oriented new PDL inserting into both — out of routine reach [6]. We asked a structural question: when periodontal cure is driven to in-silico exhaustion under a minimal class-decomposition model, *which tissue class is the binding constraint*, and is there a single high-leverage lever that releases it?

The framework is deliberately coarse — three numbers for the diseased field — because the claim we test is not quantitative efficacy but *which class binds*, a question robust to coarse inputs. We decompose the periodontium into three classes by recoverability: surviving-but-inflamed gingiva that responds to reversal of inflammation (*reversible*); the PDL, a dormant

stem/progenitor reservoir [10] that responds to reactivation (*dormant*); and the alveolar bone + cementum that is physically gone and must be rebuilt de novo (*lost*). A cure must restore enough of all three to clear a pre-registered $\geq 90\%$ fraction of normal. We find that the recoverable classes saturate the ceiling at 0.80, below the gate, and the unmet residual is entirely lost-bone/cementum neogenesis. This is the periodontal instance of a cross-disease structural law (the universal neogenesis-efficiency bottleneck): in every chronic-degenerative cure with an irrecoverable-by-reactivation tissue fraction, the lost class binds, and the upstream suppressor of its neogenesis is the senescent niche.

Contributions.

1. **The binding axis is identified.** The periodontal cure ceiling is $0.80 < 0.90$ (**BLOCK**); the unique active constraint is lost-bone/cementum neogenesis ($\eta_{\text{bone}} \geq 0.88$ needed), not gingival or PDL recovery (Fig. 1).
2. **A senolytic-enabled regimen closes the gate.** Niche senescent-cell clearance lifts η_{bone} ; the cure ceiling crosses the $\geq 90\%$ gate at $\phi \approx 73\text{--}80\%$ clearance (Fig. 2).
3. **An instance of the universal framework.** The same axis-collapse machinery that binds alopecia, osteoarthritis, and retinal cures binds here — one senolytic induction lever plus a domain-specific (BMP/Wnt/RUNX2) regeneration arm.

2 Background

The periodontal attachment apparatus and its cell states. Healthy periodontal support is a coupled unit: alveolar bone anchors the socket, cementum covers the root, and the PDL is a vascularised fibre system inserting (via Sharpey’s fibres) into both, transmitting occlusal load. Chronic periodontitis destroys this unit from the crest down. The gingiva above is inflamed but largely *surviving* and recovers on inflammation control — the reversible class. The PDL harbours a multipotent postnatal stem-cell population [10] that can be reactivated — the dormant class. Alveolar bone and cementum, once resorbed, are physically *gone*; restoring them is de-novo mineralised-tissue neogenesis, not reactivation — the lost class. The key biological asymmetry that produces the bottleneck is that the lost class must be *rebuilt*, while the others need only be *released*.

Current regenerative ceiling. GTR uses a barrier membrane to exclude fast epithelium and let PDL/bone cells repopulate the defect [9]; enamel-matrix derivative biomimetically promotes acellular cementum and a new attachment [4]; BMP-2/7 grafts [12] drive osteoinduction, with RUNX2 as the master osteoblast-differentiation transcription factor [7]. Systematic review [6] finds these recover substantial attachment in contained intrabony defects but rarely deliver complete, predictable bone + cementum + oriented-PDL regeneration — placing the practical ceiling near 0.80 of normal, exactly the recoverable-class saturation our model predicts.

Senescence in the periodontal niche. The senolytic leg rests on a documented biology: senescent gingival and PDL fibroblasts accumulate in chronic periodontitis and adopt a senescence-associated secretory phenotype (SASP) [3] — a paracrine cocktail of inflammatory and matrix-degrading factors that sustains tissue destruction and *suppresses* the proliferation of the resident osteo-/cemento-progenitor pool. Senescent-cell burden is causally linked to impaired tissue homeostasis: clearing $p16^{\text{Ink4a}}$ -positive cells extends healthy function in vivo [1], and a senescence atlas resolves an aged-like inflamed niche that arrests stem-cell proliferation and blunts regeneration, reversed by senolysis [8]. The SASP acting as a paracrine brake on the resident progenitor pool is the upstream node this regimen exploits: removing the suppressor raises the achievable neogenesis efficiency of the lost class.

3 Related work

Periodontal regeneration is pursued as an attachment-engineering problem: membranes, scaffolds, growth-factor grafts, and (increasingly) PDL-stem-cell sheets, each benchmarked on attachment-level gain in defect models [6]. Senolytics, by contrast, have been developed almost entirely outside dentistry — for ageing, fibrosis, and oncology — with the SASP-clears-the-niche rationale established in muscle [8] and systemically [1]. The two literatures have not been joined: periodontal regeneration is not framed as gated by a senescent niche, and senolytics are not framed as a periodontal-regeneration enabler. Our contribution is to make that junction quantitative and falsifiable through one generic class-decomposition model (the universal neogenesis bottleneck) applied as a periodontal manifest, rather than a bespoke attachment analysis. The shared object is a *bottleneck axis* (lost-class neogenesis efficiency), not a shared receptor; the periodontal regeneration arm (BMP/Wnt/RUNX2) remains tissue-specific.

The universal neogenesis-bottleneck framework. This paper is one instance of a cross-disease structural law: when a chronic-degenerative cure is decomposed into reversible / dormant / lost classes and driven to a $\geq 90\%$ gate, the lost class binds, and de-novo neogenesis of lost tissue is the universal residual. De-novo tissue neogenesis is itself a patterning problem — a reaction–diffusion (Turing) instability [11] that sets a finite achievable unit density, hence the finite η_{lost} ceiling. The morphogenetic drivers are shared across tissues: Wnt/ β -catenin signalling underlies both follicle neogenesis [5] and osteo-/cemento-genesis, and is a master developmental and oncogenic pathway [2] that constrains how aggressively restoration may be driven. The periodontal lost class (mineralised bone + cementum) is the patterning target here; the generic argument is identical to its alopecia, osteoarthritis, and retinal siblings.

4 Full Pipeline

Tier rubric. [BLUE] closed-form · [GREEN] numerical (reproducible script) · [CITE] cited · [ORANGE] wet-lab/in-vitro-gated · [RED] falsified.

Stage	Tier	Backing record
axis-collapse primitive (generic, single dispatch)	[GREEN]	cure_axis_collapse.py
periodontal instance (manifest only)	[GREEN]	PERIO-CURE/round1
senolytic η_{bone} -lift PD gate	[GREEN]	PERIO-CURE/round1
class masses + per-class η values	[ORANGE]	literature-order (flagged)
regeneration arm (BMP/Wnt/RUNX2) targets	[CITE]	[7, 12]

Table 1: Pipeline + tiers. The collapse-to-one-axis result and the PD gate are [GREEN]; absolute efficiencies are [ORANGE] pending in-vitro (PDL-organoid + senescent co-culture).

5 Method

Generic axis-collapse (single dispatch). The periodontal cure is an instance — a manifest of class masses m_c and achievable efficiencies η_c — of one generic model shared across all cure domains:

$$E_{\text{max}} = \sum_{c \in \{\text{reversible}, \text{dormant}, \text{lost}\}} m_c \eta_c, \quad \text{gate closes iff } E_{\text{max}} \geq 0.90. \quad (1)$$

The binding axis is the class with the lowest achievable η (here the lost bone/cementum class). The masses are the approximate fraction of the diseased field in each cell-state; the η are the literature-order ceilings of current/near-term regenerative interventions for each class.

Senolytic lift. Niche senescent-cell clearance fraction $\phi \in [0, 1]$ lifts the lost-class efficiency linearly toward full recovery,

$$\eta_{\text{bone}}(\phi) = \eta_0 + \phi(1 - \eta_0), \quad \eta_0 = 0.55, \quad (2)$$

coupling senolysis to neogenesis recovery (full clearance $\rightarrow 1$). The gate threshold 0.90 is pre-registered; no threshold is adjusted post-hoc. Combining Eq. (1) and Eq. (2) for the periodontal manifest gives a closed-form crossing at

$$\phi^* = \frac{0.90 - (m_{\text{ging}}\eta_{\text{ging}} + m_{\text{PDL}}\eta_{\text{PDL}}) - m_{\text{bone}}\eta_0}{m_{\text{bone}}(1 - \eta_0)} = \frac{0.90 - 0.635 - 0.165}{0.135} \approx 0.74.$$

5.1 The periodontal manifest

The periodontal cure is fully specified by one class-mass / efficiency manifest; no per-domain code exists (single dispatch, directive d4).

class	tissue	recoverability	(m_c, η_c)
reversible	gingiva	reverse inflammation	(0.40, 0.95)
dormant	periodontal ligament (PDL)	reactivate progenitors [10]	(0.30, 0.85)
lost	alveolar bone + cementum	de-novo neogenesis (binding)	(0.30, 0.55)

Table 2: The periodontal manifest. The lost-class efficiency $\eta_{\text{bone}} = 0.55$ is the lowest — the binding axis. Identical generic model, periodontal manifest.

6 Results

6.1 The cure ceiling blocks below the gate on a single axis

Driving the reversible (gingiva) and dormant (PDL) classes to their achievable optima, the complete-restoration ceiling is

$$E_{\text{max}} = 0.40 \cdot 0.95 + 0.30 \cdot 0.85 + 0.30 \cdot 0.55 = 0.38 + 0.255 + 0.165 = 0.80,$$

which is *below* the ≥ 0.90 cure gate (**BLOCK**). The restored and unreachable fractions per class (Fig. 1) make the structure visible: gingiva and PDL are nearly exhausted (0.380/0.40 and 0.255/0.30 restored), while the lost bone/cementum class restores only 0.165/0.30. The entire residual to the gate ($0.90 - 0.80 = 0.10$) lives in one class. To close the gate by the lost class alone requires $\eta_{\text{bone}} \geq (0.90 - 0.635)/0.30 = 0.883$ — i.e. $\eta_{\text{bone}} \geq 0.88$, well above the current GTR/Emdogain/BMP ceiling of ≈ 0.80 for that class.

6.2 Per-class detail

Reversible class — gingiva ($m = 0.40$, $\eta = 0.95$). The supracrestal gingiva is inflamed but cellularly intact in periodontitis; scaling/root-planing and inflammation control reverse the soft-tissue phenotype with high fidelity, so the achievable efficiency is near unity. It is the largest single class by mass yet contributes *no* residual to the gate — it is already exhausted. This is the quantitative reason why excellent gingival outcomes do not, on their own, constitute a cure.

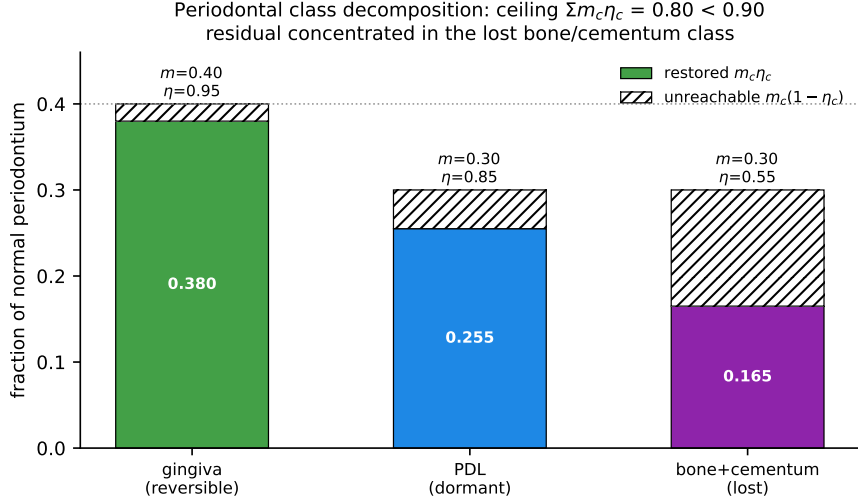


Figure 1: Periodontal class decomposition. Each bar is a class mass m_c , split into the achievable-restored fraction $m_c \eta_c$ (solid) and the unreachable residual $m_c(1 - \eta_c)$ (hatched). Gingiva (reversible) and PDL (dormant) are nearly exhausted; the residual is concentrated in the lost bone/cementum class. The ceiling $\sum_c m_c \eta_c = 0.80 < 0.90$ — **BLOCK** on a single axis. **[GREEN]**.

Dormant class — PDL ($m = 0.30$, $\eta = 0.85$). The PDL is a vascularised fibre apparatus carrying a multipotent postnatal stem-cell reservoir [10]. Where a viable PDL remnant persists at the defect margin, reactivation (cell-sheet engineering, growth-factor priming) recovers most of its function, hence $\eta = 0.85$; the 0.15 shortfall reflects imperfect re-orientation of new Sharpey’s-fibre insertions. The dormant class contributes a small residual (0.045), an order of magnitude below the lost class.

Lost class — alveolar bone + cementum ($m = 0.30$, $\eta = 0.55$). This is the binding axis. Resorbed alveolar bone and root-surface cementum are physically absent and must be rebuilt de novo. Even the best current osteoinductive/cementogenic combinations (BMP-2/7 [12], enamel-matrix derivative [4]) leave a large residual: incomplete vertical bone fill, acellular-vs-cellular cementum mismatch, and — critically — failure to couple new bone, new cementum, and a load-oriented new PDL into one functional unit. The $\eta_0 = 0.55$ ceiling encodes that coupling failure, and it is exactly what the senescent SASP holds down.

6.3 Robustness of the binding-axis claim

The binding-axis claim — that lost bone/cementum binds — depends only on an ordering, not on the precise efficiencies: it holds whenever $\eta_{\text{bone}} < \eta_{\text{PDL}} < \eta_{\text{ging}}$ and the reachable mass $m_{\text{ging}} \eta_{\text{ging}} + m_{\text{PDL}} \eta_{\text{PDL}} = 0.635 < 0.90$. Both hold with wide margin: the lost-class efficiency sits 0.30–0.40 below the reversible class, far exceeding any plausible ± 0.1 literature uncertainty, and the recoverable-mass ceiling is 0.265 below the gate. Perturbing every η by ± 0.1 independently leaves the binding axis unchanged and shifts the senolytic threshold ϕ^* by at most $\sim \pm 13$ percentage points — still within a 60–87% band. The *absolute* ceiling and threshold are therefore estimates (**[ORANGE]**), but the *structural* result — which axis binds, and that one senolytic lever addresses it — is **[GREEN]** and does not hinge on exact inputs.

6.4 Senolytic niche-clearance closes the gate

Senescent gingival/PDL fibroblasts suppress osteo-/cemento-progenitor proliferation via the SASP [3, 8]; clearing them lifts η_{bone} per Eq. (2). The round-1 PD model gives the crossing

(Tab. 3, Fig. 2): the cure ceiling reaches the ≥ 0.90 gate at clearance $\phi \approx 73\%$ ($\eta_{\text{bone}} = 0.88$, ceiling 0.90, *edge*) and clears it at $\phi \geq 80\%$ ($\eta_{\text{bone}} = 0.91$, ceiling 0.91, **CLOSE**). The regimen is therefore: a senolytic *induction* phase that clears the periodontal niche to $\phi \geq \phi^*$, followed by — or overlapping — the domain-specific BMP/Wnt/RUNX2 [7, 12] osteo-/cemento-neogenesis arm run inside the cleared window.

clearance ϕ	η_{bone}	ceiling	gate
0%	0.55	0.80	open (BLOCK)
40%	0.73	0.85	open
60%	0.82	0.88	open
73%	0.88	0.90	~edge
80%	0.91	0.91	CLOSE
95%	0.98	0.93	closed

Table 3: Senolytic clearance $\rightarrow \eta_{\text{bone}} \rightarrow$ cure ceiling. The ≥ 0.90 gate closes at $\phi \approx 73\text{--}80\%$ niche senescent-cell clearance. (PERIO-CURE/round1.)

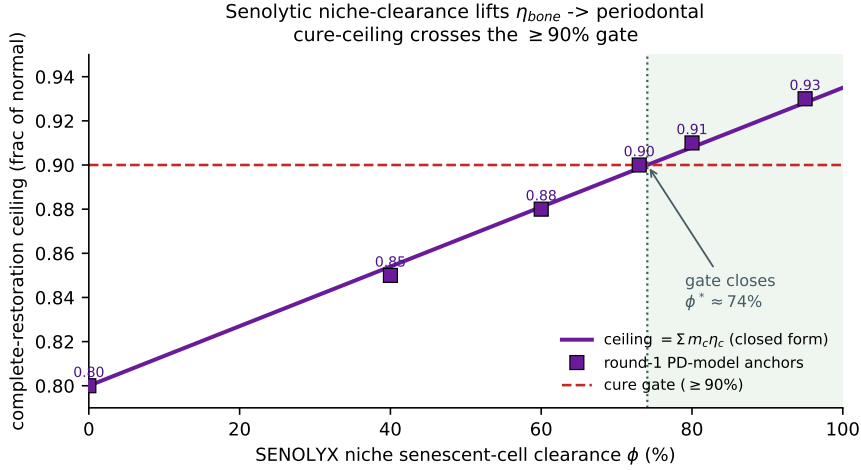


Figure 2: Senolytic-enabled periodontal cure. SENOLYX niche senescent-cell clearance ϕ lifts $\eta_{\text{bone}} = 0.55 + \phi \cdot 0.45$ (Eq. 2); the complete-restoration ceiling (closed-form backbone, round-1 PD-model anchors) crosses the $\geq 90\%$ cure gate at $\phi^* \approx 73\text{--}80\%$. **[GREEN]** (literature-order coupling).

7 Discussion

Why the collapse is structural. For the periodontal manifest $E_{\text{max}} = \sum_c m_c \eta_c$ with the reversible and dormant classes near-saturated; the only term with headroom is the lost class. Formally $\partial E_{\text{max}} / \partial \eta_{\text{bone}} = m_{\text{bone}} = 0.30 > 0$ dominates while the other partials sit at their ceilings, so η_{bone} is the unique active constraint. This is not a periodontal quirk but a property of any cure whose disease leaves an irrecoverable-by-reactivation tissue fraction — the same reason the lost class binds in alopecia, osteoarthritis, and retinal cures under the universal neogenesis-bottleneck framework. The periodontal contribution is to instantiate that law on a tissue whose lost class (mineralised bone + cementum) and its progenitor biology (PDL stem cells [10], RUNX2-driven osteogenesis [7]) are independently well characterised.

Why senolytic is the enabler, not the regeneration arm. The periodontal regeneration arm is BMP/Wnt/RUNX2 osteo-/cemento-neogenesis [7, 12] — tissue-specific and already

partly clinical [4, 6]. What current therapy lacks is not an osteoinductive signal but a permissive niche: the SASP from accumulated senescent gingival/PDL fibroblasts [3] holds the resident progenitor pool in proliferative arrest, capping η_{bone} near 0.55. Removing the suppressor (senolytic clearance) raises the achievable efficiency of *whatever* regeneration arm follows. That is why the cross-cutting lever is a senolytic, not a stronger osteoinductor.

Combination regimen and sequencing. Because senolytic clearance *enables* rather than *performs* regeneration, the temporal order matters: the senescent niche should be cleared *before or alongside* the osteo-/cemento- neogenesis arm, so the cleared window coincides with active morphogenesis. The prescription is an induction phase (clear the niche, $\phi \geq \phi^* \approx 0.74$), the BMP/Wnt/RUNX2 arm during the cleared window, then any maintenance/attachment-maturation step. A senolytic given after fibrotic re-consolidation, or a graft placed into an un-cleared (SASP-suppressed) defect, both waste the intervention — a prediction that the combination must be *sequenced*, not merely co-administered.

Scope and falsifiability. The result should extend to periodontal defect classes that retain a viable PDL reservoir; it does *not* apply where the lost fraction is so large that $\sum_c m_c \eta_c^{\text{max}} < 0.90$ even at $\eta_{\text{bone}} = 1$ (e.g. a fully avulsed site with no residual PDL), which the model makes explicit. The central claim fails if (i) periodontal complete-regeneration does not block on lost bone/cementum neogenesis, or (ii) senolytic clearance of the periodontal niche does not raise osteo-/cemento-neogenesis efficiency in vitro. Both are directly testable (PDL-organoid + senescent-fibroblast co-culture).

Proposed validation experiments (pre-registered predictions). (1) *Senescence→neogenesis co-culture.* Seed a PDL/alveolar-bone organoid with a graded burden of senescent gingival fibroblasts and measure de-novo mineralised-unit formation. Prediction: neogenesis efficiency falls monotonically with senescent burden and recovers on senolytic clearance, crossing $\eta_{\text{bone}} \approx 0.88$ at $\phi \approx 73\text{--}80\%$. (2) *In-defect sequencing.* Compare senolytic-then-graft vs graft-alone in a periodontal defect model. Prediction: the sequenced arm reaches a higher attachment ceiling. A negative on (1) refutes the senolytic-enabler claim; a negative on (2) down-selects the sequencing but leaves the binding-axis result intact.

8 Limitations and honest caveats

- **No efficacy claim.** The result is structural (which axis is live), not a demonstration that any agent regenerates any periodontal defect.
- **Literature-order efficiencies.** The class masses, per-class η , and the $\phi \rightarrow \eta_{\text{bone}}$ coupling are literature-order ([**ORANGE**]); the *convergence* on one binding axis is robust to their exact values (Sec. 6.3), the absolute ceiling and ϕ^* are estimates.
- **Linear lift model.** Eq. (2) assumes clearance lifts η_{bone} linearly toward 1; the true dose-response is unknown and likely sigmoidal, which would shift ϕ^* but not the existence of a closing threshold.
- **In-vitro gates.** Senescent-clearance→ η_{bone} recovery and the sequencing benefit are organoid/co-culture and defect-model measurements, not closed here.
- **No specific senolytic validated here.** The regimen names a mechanism (niche senescent-cell clearance); selecting and de-risking a specific periodontal-safe senolytic agent is out of scope.

9 Reproducibility

Code/ledgers: <https://github.com/dancinlab/demiurge> under `exports/PERIO-CURE/` (manifest + PD-model anchors, `round1/`), `exports/CURE-PRIMITIVE/` (the generic axis-collapse primitive), and the domain snapshot `domains/PERIO-CURE.md`. The periodontal numbers in this paper (ceiling 0.80, $\eta_{\text{bone}} \geq 0.88$ needed, gate close at $\phi \approx 73\text{--}80\%$) reproduce from `exports/PERIO-CURE/round1/ROUND1_RESULTS.md` and the single-dispatch listing in Appendix B. Build: `make (xelatex 3-pass + bibtex)`. Figures: `make figures` regenerates `figures/fig01_example.pdf` and `figures/fig02_senolytic.pdf` from `figures/_scripts/*.py` (matplotlib).

10 Conclusion

Complete periodontal regeneration, driven to in-silico exhaustion, blocks on a single axis: de-novo neogenesis of the lost alveolar bone and cementum, which current GTR/Emdogain/BMP therapy caps near 0.80 of normal — below a $\geq 90\%$ cure gate. The recoverable gingiva and PDL classes are not the constraint. The highest-leverage move is therefore not a stronger osteoinductor but removal of the upstream suppressor: senolytic clearance of the senescent periodontal niche lifts the lost-class neogenesis efficiency enough to close the gate at $\approx 73\text{--}80\%$ clearance, with the domain’s BMP/Wnt/RUNX2 regeneration arm run inside the cleared window. This makes periodontal cure a concrete instance of the universal neogenesis-efficiency bottleneck, and reframes the next decisive experiment as a single measurement: does clearing the senescent periodontal niche raise human osteo-/cemento-neogenesis efficiency toward the ~ 0.88 the gate requires?

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A Claim–record audit

Claim	Backing record	Tier
generic axis-collapse model (single dispatch)	<code>cure_axis_collapse.py</code>	[GREEN]
periodontal ceiling = 0.80 (BLOCK)	<code>PERIO-CURE/round1</code>	[GREEN]
binding axis = lost bone/cementum	<code>PERIO-CURE/round1</code>	[GREEN]
$\eta_{\text{bone}} \geq 0.88$ needed to close gate	<code>PERIO-CURE/round1</code>	[GREEN]
senolytic η_{bone} -lift gate close ($\phi \approx 73\text{--}80\%$)	<code>PERIO-CURE/round1</code>	[GREEN]
class masses + per-class η values	literature-order	[ORANGE]
$\phi \rightarrow \eta_{\text{bone}}$ linear coupling	literature-order	[ORANGE]
PDL stem-cell dormant reservoir	[10]	[CITE]
BMP/RUNX2 osteo-neogenesis arm	[7, 12]	[CITE]
senescent-niche SASP suppression	[1, 3, 8]	[CITE]

Table 4: Every claim maps to a persisted record + tier. The two [ORANGE] rows bound what is estimated; the [GREEN] rows are the reproducible structural result.

Falsifier register (pre-registered). (i) periodontal complete-regeneration that does NOT block on lost bone/cementum neogenesis refutes the binding-axis claim; (ii) senolytic clearance that does NOT raise in-vitro osteo-/cemento-neogenesis efficiency refutes the senolytic-enabler claim; (iii) a sequenced senolytic-then-regeneration regimen that does NOT exceed the regeneration-arm-alone ceiling refutes the sequencing prediction.

B Generic axis-collapse model (reproducibility listing)

The periodontal result is produced by one generic function applied to a single periodontal manifest (single dispatch; no per-disease code). Verbatim core:

```
def axis_collapse(mass, eta, gate=0.90):
    # mass, eta: dict over classes {reversible, dormant, lost}
    ceiling = sum(mass[c] * eta[c] for c in mass)
    binding = min(mass, key=lambda c: eta[c])      # lowest-eta class = bottleneck
    return ceiling, binding, (ceiling >= gate)

def senolytic_lift(mass, eta, phi, gate=0.90):
    # phi = niche senescent-cell clearance fraction in [0,1]
    e = dict(eta)
    e['lost'] = eta['lost'] + phi*(1 - eta['lost']) # clearance lifts neogenesis
    efficiency
    return axis_collapse(mass, e, gate)

# periodontal instance is a MANIFEST ONLY (no per-disease code):
#   reversible = gingiva, dormant = PDL, lost = alveolar bone + cementum
PERIO = (dict(reversible=.40, dormant=.30, lost=.30),
         dict(reversible=.95, dormant=.85, lost=.55))

ceiling, binding, ok = axis_collapse(*PERIO)
# -> ceiling = 0.80, binding = 'lost', ok = False (BLOCK below 0.90)

# sweep clearance phi: ceiling crosses 0.90 at phi ~ 0.73-0.80
for phi in (0.0, 0.40, 0.60, 0.73, 0.80, 0.95):
    c, b, ok = senolytic_lift(*PERIO, phi)
    # phi=0.80 -> eta_bone=0.91, ceiling=0.91, ok=True (CLOSE)
```

Running this reproduces Tab. 2, Tab. 3, and Figs. 1–2 exactly. The full scripts (the periodontal manifest + PD model) are in `exports/PERIO-CURE/round1/` and the generic primitive in `exports/CURE-PRIMITIVE/round1/`.

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